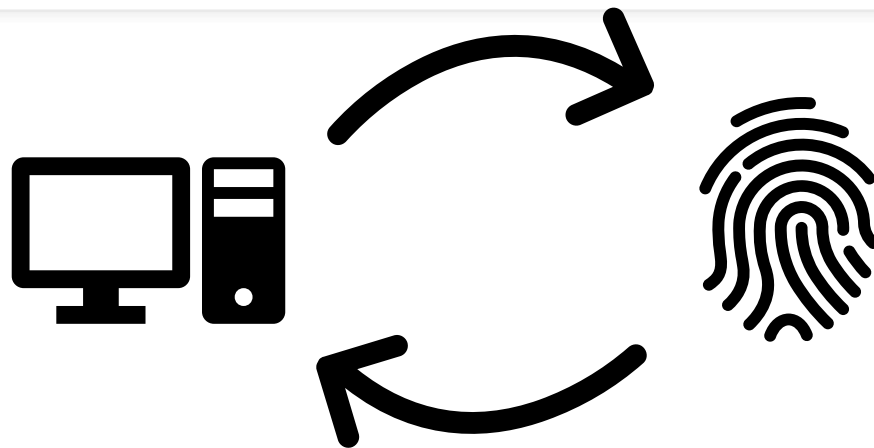
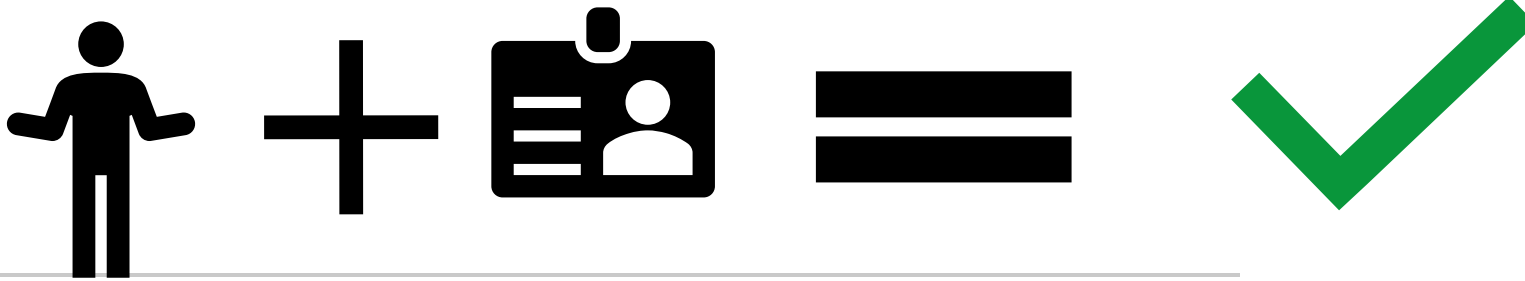


Project II. Instrumented & Monitored Cloud Service



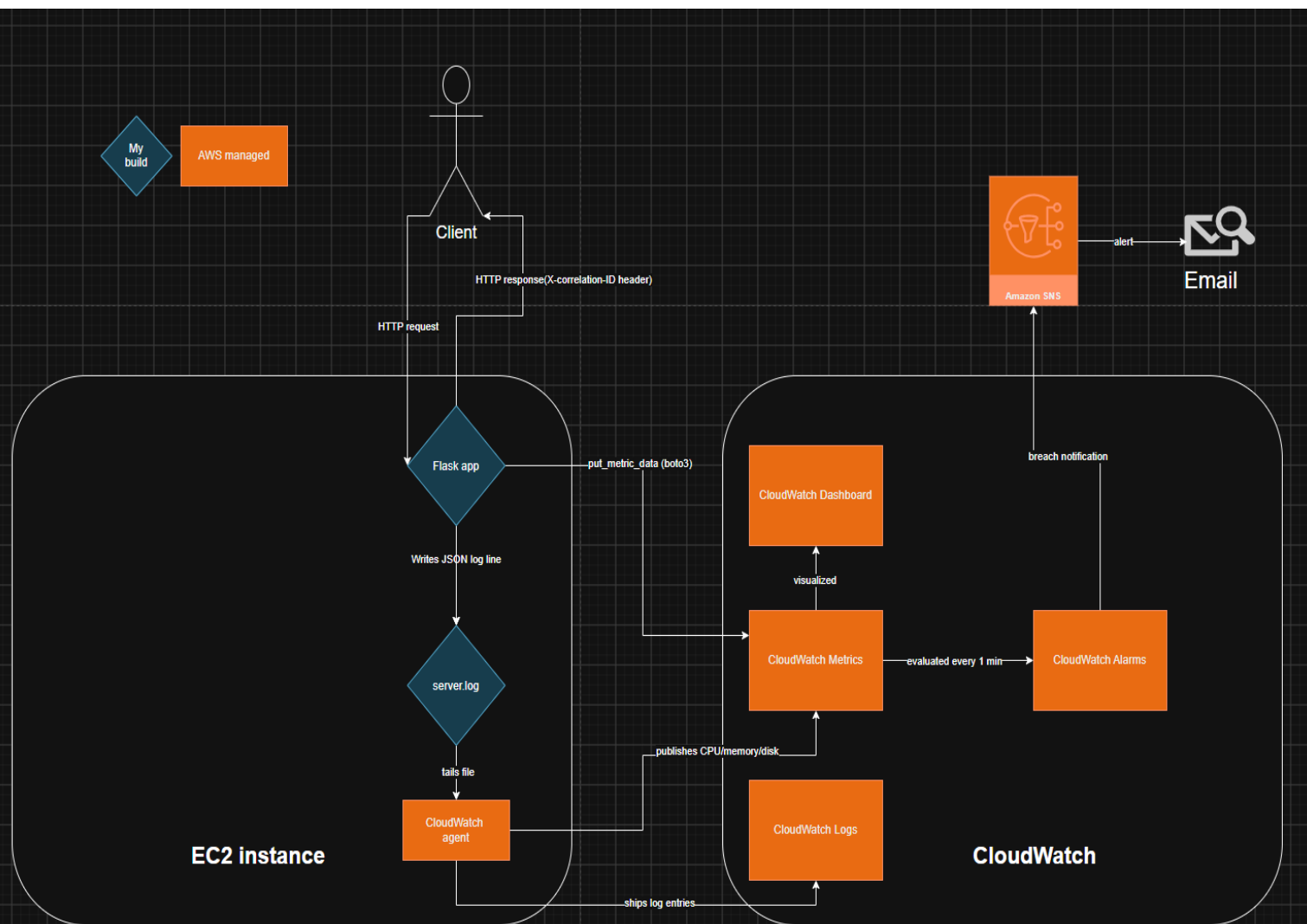
The Application



A small authentication service

- Register
- Login / Logout
- Suspend accounts
- Saves accounts in memory

Architecture



- Simple EC2 instance
- Python Flask + Gunicorn
- Single boto3 worker
- Agent tails JSON log file
- Every request gets correlation ID
- 3 log levels (info, warning, error)

Live Demo

CloudWatch
dashboard
&
Log files

What Happened?

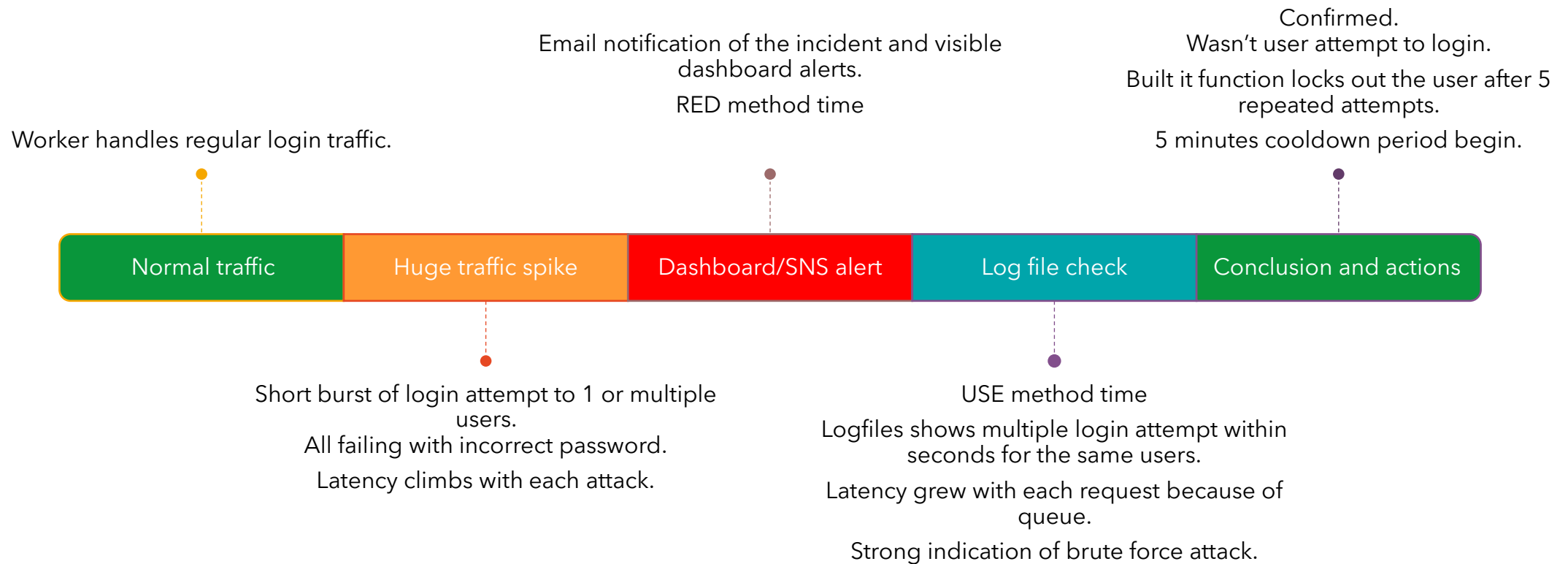
1. A downstream dependency failure, like the database timing out.
2. A legitimate traffic surge, a marketing push or a viral post.
3. A recent bad deploy causing login failures across many accounts.
4. A brute-force attack against one or a few specific accounts.
5. A false alarm, a monitoring or metrics glitch, not a real incident.

Choose the correct answer

Incident Response

Simulation

Timeline



Alerting and Response

2 warnings:

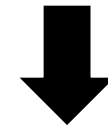
Warning at 10 failed / min
Critical at 25 failed / min

The Lockout:

Threshold of 1

2 latencies:

Warning at 500ms
Critical at 1000ms



Baseline is
~300ms

2 separate SNS topic built so they never look identical.

Learning & Improvement

What I learnt

- Point-in-time alarms check can't tell it all
- A short burst beats short evaluation period
- SNS subscription as single point of failure
- No DB + multiple worker = inconsistent data
- Detection and delivery are two separate systems

Future improvements

- Check logic + test before going further
- More notification channels
- Make it scalable with DB
- Notification for user to take actions
- IP ban to prevent further attack
- Add canary check to confirm notification delivery

Q&A





Thank you for the attention

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